

# Ryley Scott Crow

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## Education

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George Washington University, Washington, DC Fall 2023 - Present

- Ph.D. Candidate, Biological Sciences, expected 2028. Advisor: Dr. Leon Grayfer.
- NSF Graduate Research Fellow.

University of Virginia, Charlottesville, VA - 3.93 GPA Fall 2018 - Spring 2021

- B.A. Biology and Environmental Sciences, with Highest Distinction.
- Honors - Phi Beta Kappa, Dean's List, Distinguished Majors Program, Harrison Grant Recipient, Wolfe Undergraduate Fellowship.
- Thesis - "Marine heatwave effects on the germination and molecular response of eelgrass (*Zostera marina*)".

## Research Experience

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Ph.D. Candidate, George Washington University: Grayfer Lab, Washington, DC Fall 2023 - Present

- Study mast cells in the amphibian *Xenopus laevis* to define conserved and unique features of amphibian and mammalian mast cells.
- Characterize mast cell activation pathways, including IgY-mediated degranulation and antibody-independent activation through MRGPRs, venom peptides, and pathogen recognition.
- Compare peritoneal- and bone marrow-derived mast cell cultures using RNAseq, proteomics, cytology, and degranulation assays.
- Investigate the roles of skin mast cells in defense against the chytrid fungus *Batrachochytrium dendrobatidis* (Bd).

Lab Technician, George Washington University: Smith Lab, Washington, DC June 2021 - Fall 2023

- Evaluated the novel SpTransformer (SpTrf) proteins and their function in the immune system of *Strongylocentrotus purpuratus*.
- Assessed the opsonin capacity of the SpTrf proteins by evaluating the phagocytosis of micron spheres crosslinked to SpTrf proteins.
- Demonstrated that soluble SpTrf proteins bind phagocytes and modulate SpTrf gene expression and immune responsiveness.

Research Assistant, University of Virginia McGlathery Lab, Charlottesville, VA Fall 2020 - Summer 2021

- Researched the response of *Zostera marina* seeds to an artificial heatwave scenario for the Environmental Sciences Distinguished Majors program.
- Evaluated the expression of heat shock proteins using qPCR, as well as seed germination.

REU Intern, Friday Harbor Laboratories, San Juan Island, WA Summer 2020

- Designed a field and lab-based project to study the burial of *Zostera marina* seeds by the lugworm species *Abarenicola pacifica*.
- Analyzed experimental data using GLM (negative binomial regression) and GIS methods.

Research Assistant, University of Virginia Timko Lab, Charlottesville, VA Spring 2019 - Spring 2020

- Evaluated transcription factors on *Nicotiana* trichome development using agrobacterium transformation.

## Publications

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- **Crow, R. S.**, Grubb, C., Meisel, J. W., and L. C. Smith (2026). "Recombinant SpTrf proteins illustrate putative functions of native SpTrf proteins: from immune response to resolution in sea urchins". *The Journal of Immunology*, 215(8).

- **Crow, R.**, Tesega, H., Chapkin, E. R., Garvey Griffith, C. N., Hossainey, M. R. H., Ma, J., and L. Grayfer (2026). "A comparison of amphibian (*Xenopus laevis*) peritoneal- and bone marrow-derived mast cells". *The Journal of Immunology*, 215(2).
- **Crow, R. S.**, Grayfer, L., and L. C. Smith (2025). "Recombinant SpTransformer proteins bind to specific sites on sea urchin phagocytes and modulate SpTransformer gene expression and immune responsiveness". *Frontiers in Immunology*, 15:1496832.
- Hauser, K. A., Garvey, C. N., **Crow, R. S.**, Hossainey, M. R. H., Howard, D. T., Ranganathan, N., Gentry, L. K., Yaparla, A., Kalia, N., Zelle, M., Jones, E. J., Duttargi, A. N., Rollins-Smith, L. A., Muletz-Wolz, C. R., and L. Grayfer (2024). "Amphibian mast cells serve as barriers to chytrid fungus infections". *eLife*, 12:RP92168.
- **Crow, R. S.**, Shaw, C. G., Grayfer, L., and L. C. Smith (2024). "Recombinant SpTransformer proteins are functionally diverse for binding and phagocytosis by three subtypes of sea urchin phagocytes". *Frontiers in Immunology*, 15:1372904.
- Shaw, C. G., Pavludi, C., **Crow, R. S.**, Saw, J. H., and L. C. Smith (2024). "Spotting disease disrupts the microbiome of infected purple sea urchins, *Strongylocentrotus purpuratus*". *BMC Microbiology*.
- Shaw, C. G., Pavludi, C., Barela Hudgell, M. A., **Crow, R. S.**, Saw, J. H., Pyron, R. A., and L. C. Smith (2023). "Bald sea urchin disease shifts the surface microbiome on purple sea urchins in an aquarium". *Pathogens and Disease*.
- Smith, L. C., **Crow, R. S.**, Franchi, N., and C. S. Schrankel (2023). "The echinoid complement system inferred from genome sequence searches". *Developmental and Comparative Immunology*.
- **Crow, R. S.**, Merz, R., Dethier, M., and S. Wyllie-Echeverria (2023). "*Abarenicola pacifica* burrowing behavior and its implications for *Zostera marina* seed burial, restoration, and expansion". *Pacific Science*.

## Presentations

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- **Crow, R.** (talk). "Mast cell driven susceptibility in the Panamanian golden frog", Global Amphibian and Reptile Disease (GARD) Conference, May 2026.
- **Crow, R.** (talk). "Uncovering the hallmarks of amphibian mast cell activation", North American Comparative Immunology Workshop, June 2024. Awarded 1st prize for talk.
- **Crow, R.** (talk). "SpTransformer proteins bind to the phagocyte class of coelomocytes and change gene expression in the innate immune system of the purple sea urchin", International Society of Developmental and Comparative Immunology (ISDCI) Congress, Wageningen University, Wageningen, The Netherlands, August 2023. Awarded second best talk.
- **Crow, R.**, Shaw, C. G., and Smith, L. C. (talk). "Recombinant SpTransformer proteins augment phagocytosis by sea urchin coelomocytes", North American Comparative Immunology Workshop, June 2022.
- **Crow, R.**, Ewers-Lewis, C., and McGlathery, K. (talk). "Marine heatwave effects on the germination and molecular response of eelgrass (*Zostera marina*)", Coastal and Estuarine Research Federation, October 2021. Awarded best undergraduate talk.
- **Crow, R.**, Dethier, M., and Wyllie-Echeverria, S. (poster). "Experimental evaluation of *Abarenicola pacifica* burrowing behavior", Western Society of Naturalists, Nov. 2020; Society for Integrative and Comparative Biology, Jan. 2021.

## Awards, Honors, and Grants

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### NSF Graduate Research Fellowship

2023 - Present

- Awarded the National Science Foundation Graduate Research Fellowship to support doctoral research.

### Best Undergrad Talk, CERF 2021

Fall 2021

- Awarded best talk given by an undergraduate at the Coastal and Estuarine Research Federation (CERF) conference.

Harrison Undergraduate Research Award, Charlottesville, VA

Spring 2020 - Spring 2021

- Received funding for “Marine heatwave effects on the germination and molecular response of eelgrass (*Zostera marina*)”.

Wolfe Undergraduate Fellow for “Extinction in the Archives”, Charlottesville, VA

Fall 2018 - Spring 2019

- Analyzed archival materials for the purpose of curating a display on global historic biodiversity loss and extinction.

## **Skills**

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- Primary cell culture: amphibian mast cell cultures from peritoneal and bone marrow precursors.
- Flow cytometry, immunocytochemistry, immunohistochemistry, confocal and fluorescence microscopy.
- Molecular biology: qPCR, DNA/RNA extraction, Western blot, ELISA, recombinant protein production, cloning.
- Transcriptomic (RNAseq) and proteomic data analysis; R and Matlab programming.
- Scanning electron microscopy; Geographic Information Systems (ArcGIS Pro).

## **Mentoring and Teaching Experience**

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Hamerenoah Tesega, George Washington University

Spring 2024 - Fall 2024

- Mentored Harlan Undergraduate Research on IL-4 mediated effects on amphibian mast cells.
- Awarded best poster for this work.

Athanasios Naum, George Washington University

Spring 2022 - 2023

- Undergraduate thesis. Awarded the competitive Luther Rice Undergraduate Research Fellowship.

Archi Patel, George Washington University

Summer 2022

- High school internship.

Teaching Assistant, University of Virginia

Spring 2021

- Provided teaching assistance for BIOL 3040: Developmental and Regenerative Biology.